Inferentiality Realisations and Cognitive-pragmatic Skills in Primary School Pupils' Reading Comprehension, Ogun State, Nigeria

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Abstract
Extant studies on reading comprehension strategies have largely focused on strategies such as activating prior knowledge and monitoring clarifying. However, limited attention has been given to the inferential processing and cognitive-pragmatic skills that underpin pupils' understanding of comprehension texts and cognitive. This paper, therefore, examines Nigerian pupils' ability to harness the resources of inferencing for the purpose of achieving reading comprehension from a cognitive-pragmatic perspective as well as cognitive skills' engagement. 200 pupils from 10 public primary schools in two purposively selected local government areas in Ogun State constitute the subjects for this study. A comprehension passage was administered to the pupils and their responses to this comprehension passage were analysed along the lines of the various categories of inferentiality. Relevance Theory as postulated by Sperber and Wilson (1995) with further insights from Kecskes' (2008) Socio-cognitive Approach and Van Dijk's (2009) Socio-cognitive Theory of Context serve as the theoretical anchorage for the study. The findings reveal that the categories of inferentiality – causal, evidential and temporal – are rarely engaged by the pupils in uncovering the author's intended meaning which would have necessitated an in-depth comprehension. Comprehension skills identified as lexical-semantic, pragmatic and discourse are also inadequately employed by the pupils. The paper concludes that systematic instructions and useful learners' engagement can be instrumental to entrenching effective inferencing potentials and comprehension skills in Nigerian pupils as they tackle reading comprehension tasks.
Keywords: Inference, English Language Learning, Nigerian School Pupils, Systematic instructions, comprehension skills

Introduction
Text comprehension involves constructing a coherent representation of the information contained in a text as well as those not explicitly stated. Consequently, the reader's mental representation also includes implied information that the writer expects the reader to infer from the text. Reading theorists widely agree that the ability to generate inferences is essential for effective reading comprehension (Davoudi, 2005). Inference enables readers to understand the information presented by the author and plays a critical role in both comprehension and memory for text. Skilled readers generate elaborative inferences, which can only occur when the necessary background knowledge is available. Thus, general knowledge is a key contributor to both inference generation and comprehension (Ahmed, Francis, Fletcher, Barnes, & Kulesz, 2016). When readers are unable to generate inferences that link explicit textual information to relevant world knowledge, they often experience a comprehension breakdown at one point in time or the other.

Learning to read is secondary to reading to learn which is even more fundamental in orientation. The public primary school system in Nigeria is more or less dysfunctional in terms of helping beginning readers acquire the foundational strategies necessary for comprehension to take place (Ikwen, 2014). Hence, only a small fraction of pupils who go through the Nigerian public primary school system succeed in learning to read.

Scholarly works have been done on pupils' reading comprehension. Paris and Stahl (2005) examine the processes preschool children use to understand text the events and facts they encounter and Randi, Newman and Grigorenko (2010) investigate the factors that hinder reading comprehension in children with autism. Specifically in Nigeria, Adeniji and Omale (2010) examined the teaching capabilities of selected primary school teachers in Oyo State in imparting reading comprehension skills on their pupils with a greater emphasis on reading. However, there is paucity of scholarly works on the cognitive as well as pragmatic evaluation of Nigerian pupils' ability to answer comprehension questions with emphasis on inferencing and comprehension skills and the paper intends to fill that gap. The objectives of the paper are two-folds: to investigate the application of inferentiality in Nigerian pupils' reading comprehension; and examine comprehension skills' engagement in Nigerian pupils' reading comprehension

Literature Review

Reading, listening, speaking, and writing constitute the four core language skills and can be further sub-categorised into two which are oracy and literacy skills. Listening and speaking are classified as oracy skills, whereas reading and writing are literacy skills (Enighe and Afangideh, 2018). Hence, not surprisingly, an individual is tagged a literate person in as much as he can read and write. Reading according to these scholars, is therefore one of the essential competence learners are expected to develop in the instructional language during primary education. Reading basically means making sense of printed texts. Speaking is a complex cognitive and linguistic skill and that presupposes that reading is a much more complicated cognitive and linguistic skill. Burns and Siegel (2018) submit that speaking competence entails the capability to express ideas effortlessly, precisely and suitably which lends credence to the fact that speaking is quite an engaging language activity. Human brains have been wired to speak much unlike reading which has to be consciously acquired through consistent exposure and practice although speaking also has its own peculiarities.

Permatasari, and Wulandari (2025) categorise the indices of speaking competence as fluency, accuracy and appropriateness. This could be because of a reasonable degree of the spontaneity and meticulous muscle coordination for speech production. Children usually learn to read at about the age of 5 or 6 and they actually require several years to perfect their reading skills. However, speaking comes much earlier than the age of starting to read, beginning from babbling at about 6 months to actual articulation of speech sounds and then to words leading up to full speech production much later. Speaking is natural as children naturally pick the language of their mothers and that of other people around them for some time even though learning to speak a foreign language can pose quite some difficulties. Such difficulties include mismatches of learners' learning styles or patterns of abilities and the teaching methodologies of the teachers among others (DeKeyser, 2005).

Normal children with impaired neurological system brought up through interaction with caring adults in a speech community. It then goes without saying that normal children without impaired neurological system will acquire the spoken language even more easily. In light of the foregoing, speaking is quite more easily acquired than reading. Reading is far more intentional than speaking, which is more or less automatically acquired, given the necessary conventional and linguistic environment. Reading is essential for learning in general and for acquiring languages (Hansen, 2017) and preparing the brain to decode the many complex graphic symbols in a print system is a task of education. Reading is basic to general learning because a learner needs reading skills to access the content of school subjects. It is also basic to language acquisition because it is highly necessary for vocabulary acquisition in the language being learned. A single word or phrase can be used in different contexts or situation eliciting different shades of meaning. It also enables the reader to use language with precision, fluency, accuracy and a high level of intelligibility. Reading is a challenging skill requiring prolonged practice and time to acquire (Lundahl 1998 as cited by Hansen 2017) much unlike speaking that is virtually spontaneous. Even though there are many ways of obtaining information in this present fast paced world, reading is regarded as the most valid and realistic among them all (Özyılmaz 2010:4).

The primary school level is the foundation on which an individual's readership should be built with respect to further learning assignments throughout the learner's primary schooling. Lindner (2008) highlights that a good number of pupils have weak reading abilities resulting from teachers' difficulty in developing early reading skills and limited strategy use. This implies that pupils begin to manifest reading deficiencies right from the primary schools the fact which should be a wake-up call to the teachers and other reading stakeholders to tackle the challenge right from the source. Udofot (2007) suggests extensive reading builds communicative competence by enhancing vocabulary, improving interpretative skills, and increasing reading speed. This stipulates that an individual learning to read should not restrict himself to reading only to pass tests and examinations but should be engaged in reading non-academic materials like magazines and other periodicals, novels, religious texts to mention a few and this should be recommended to our pupils right from the pre-primary section of their educational life.

Roe conceives comprehension as making meaning of what we read (2014:14). This concept of comprehension views it as a pursuit of meaning in any reading attempt. This conceptualisation is congruent with Babalola who perceives comprehension as the power to understand (1999:178). Both scholars have argued it that comprehension has a lot to do with sourcing for meaning of a text as presented by the writer. This view of comprehension sums up what comprehension entails with respect to meaning as the focal point and it also pictures comprehension as the highest achievement in reading. For comprehension to take place, a lot of both cognitive and physiological activities have to be going on simultaneously. Just and Carpenter (2013) views comprehension as involving complex cognitive processes without which an in-depth comprehension may not be achievable. One important way of viewing the cognitive processes of comprehension is that the reader constructs a mental model of the situation depicted in the text—a representation of the real or imaginary world described (Yuill and Oakhill, 1991). This mental model assists the reader in deconstructing the various concepts and events in the text for the purpose of understanding the text.

Cain, Oakhill, Barnes, and Bryant (2001) investigated the relationship between young pupils' comprehension skills and inference-making ability using a procedure that controlled for individual differences in general knowledge. They found a strong relationship between comprehension skill and inference-making ability even when knowledge was equally available to all participants. Their findings suggest that poor comprehenders construct incomplete representations of the text. Although they are often able to integrate information at a local level, they struggle to develop a coherent and integrated representation of the text as a whole, thereby limiting their ability to form an accurate mental model that supports full understanding. Difficulties with inference-making are therefore considered a likely cause of poor comprehenders' text-level comprehension problems (Cain and Oakhill, 1999). This particular work is an eye-opener into the process of deriving inference by young children when they come in contact with any comprehension text and it also enables researchers into children's comprehension to have a good grasp of how to develop better methodologies for helping young pupils master the strategies necessary for tackling difficult comprehension passages.

Halamish and Elbaz (2021) examined the effect of reading medium (screen versus paper) on pupils' reading comprehension and metacomprehension. Fifth-grade pupils read short texts, estimated their level of comprehension for each text, and subsequently completed a reading comprehension test. Two texts were read on paper, while two others were read on screen. The results indicated that pupils' reading comprehension was better when reading on paper than on screen, although the initial reading time was similar across both media. The findings were independent of medium preferences, computer usage habits, or reading ability. Furthermore, pupils' metacomprehension judgments were largely insensitive to the effect of reading medium, and their preferences suggested indifference toward the medium used, both before and after completing the task in both formats. These findings suggest that, similar to adults, children are often metacognitively unaware of the negative effect of on-screen reading on comprehension and may therefore make ineffective choices regarding the medium used for reading tasks. Studies like these are relevant in that they can assist standard examination bodies to make decisions that will favour the success tendencies of the target candidates especially with regards to the medium of the examination, either by screen or by paper.

In profiling pupils' reading comprehension, Gruhn, Segers, Keuning, and Verhoeven (2020) developed a dynamic approach with componential abilities (orthographic knowledge, vocabulary, sentence-integration) assessed within the same texts and accompanied by feedback in addition to global comprehension of these texts. In a study of 275 Dutch third- to fifth-graders, the researchers examined the extent to which response accuracy for questions on componential abilities—both on first attempts and after feedback—predicted global text comprehension within the same texts, as well as subsequent growth in a standardised reading comprehension test. They found that global text comprehension increased with each correctly answered componential question on first attempts and with each correctly answered sentence-integration question after feedback. Accuracy on first attempts also uniquely accounted for variance in growth on the standardised reading comprehension test. According to Gruhn et al., a dynamic approach may thus provide a better understanding of children's reading comprehension profiles. This research will be a big leap in assessing children's reading comprehension in

order to ascertain how globally or locally they can achieve text coherence in comprehension.

Theoretical Anchorage

Relevance theory, henceforth RT, concerns the inferential and contextual aspects of language, focusing on how what is merely implied in an utterance helps determine the meaning of what is explicitly stated. Sperber and Wilson (1995) assert that in the process of communicating, the hearer or reader, as the case may be, is guided by a single and quite generally accepted criterion to assess a variety of possible interpretations of a particular piece of communication. Insights from RT can help to shed light on the sort of interconnectivity between the writer and reader of a particular comprehension text in the sense that the intention of the writer has to be figured out by the reader. In addition to that, the reader is saddled with more responsibilities as he or she will from the same passage decipher the subject matter of the passage, that is the main message of the passage as well as what precisely the author is saying about the subject matter.

Additionally, an aspect of SCA known as the Dynamic Model of Meaning (Kecskes, 2008) also guide the analysis. DMM integrates internal and external approaches to context, viewing the communicative process as a structured whole and advocating for an encompassing perception of context that includes both prior and present experiences with the external world. It also highlights the meaning values of words, emphasising that words act as an interface connecting phonological, syntactic, lexical, and conceptual structures during comprehension and production of utterances (Culicover, Varaschin, & Winkler, 2022).

Van Dijk (2009) in his Socio-cognitive Theory of Context views context as a subjective mental model which is often constructed by participants and relevant aspects of communication situation are represented in that mental model. This theory of context is a shift in the focus of pragmatics from the external contextual factors to internal and cognitive representations of the communicative situation. Hence, contexts are subjective interpretations of the dynamically changing properties of the communicative situation as understood by participants. This paper views comprehension as context- and inference-driven and the DMM with Socio-cognitive Theory of Context as postulated by Van Dijk will cater for the context-driven aspect of comprehension.

Methodology

The data collection took place between 2022 and 2023. 200 pupils across 10 public primary schools in two purposively selected local government areas in Ogun State namely Abeokuta South and Ado-Odo Ota, 10 from each, constitute the subjects for this study. The schools were purposively selected because the local government areas are centrally located within the State and hence, representative in the study's results presentation. The pupils were in their upper primary schools, specifically Primaries 4 and 5, 100 pupils from Primary 4 and 100 pupils in Primary 5. A comprehension passage was administered to those pupils and their responses to this comprehension passage were analysed along the lines of the various categories of inferentiality identified as causal, evidential and temporal inferentialities. The passage was selected from the Nigerian Primary English, Book 4 by Ademola-Adeoye, Adam, Somoye, Omiteru, Macauley, Oderinde, Badaiki,

Hawkes and Dallas, published by LearnAfrica, in 2019. It was purposively selected because it underscores a general discourse with moral lessons which would in addition to assessing the pupils' comprehension skills, assist the pupils in inculcating proper morals as they are still in their formative years. The research design is descriptive as it seeks to describe the current reading comprehension situation among the Nigerian primary school pupils. The rationale for selecting the passage is that it is curled from a Basic 4 English textbook, making the language accessible to both Primaries 4 and 5 pupils that constituted the subjects for this study.

Analysis and Findings

Below is the comprehension passage that was administered to the pupils after which comes the analysis of the findings from the responses they supplied to the comprehension test.

One Good Turn Deserves Another

Mariam lived in Zaria. She worked at the immigration office. One day, a young man named Barry came to her office. He wanted to obtain a Nigerian passport. He had just won a scholarship to study Law in the United States of America and had to be in New York the following week. He knew how difficult it was to obtain a Nigerian passport and to make it worse, he needed the passport urgently.

Barry approached Mariam and told her of his great need. Mariam understood that this was an opportunity Barry must not miss and decided to help him. She gave him the application forms and told him to fill them. He quickly paid the application fee, then she told him to submit his application forms with some documents.

Barry submitted his forms promptly. Mariam took the application forms to her boss and pleaded with him to help Barry so that his passport would be ready within the week. Her boss wanted a bribe, but Mariam explained that Barry was poor and could not afford a bribe. After much pleading, her boss decided to help. Within a week Barry was able to obtain his passport. He thanked Mariam for her cooperation and kindness, and left for the New York.

For three years, Barry studied Law. He graduated and came back to Nigeria to attend the Law School. When he got to Nigeria, he went to the immigration office to look for Mariam, but was told she no longer worked there.

Many years later, Barry became a brilliant lawyer, known all over the country. Meanwhile, Mariam was wondering what had happened to him. She did not hear from him as he had promised, but she was patient and full of hope that they would meet again.

Barry met Mariam one day at a supermarket. There, he learned that she had been unlawfully dismissed from her office for refusing to take a bribe as directed by her boss. Barry decided that this was the time to offer his help.

Barry and Mariam decided to work together as a team. Mariam supplied him with all the information he needed and he noted all the illegal deals that took place in the immigration department. After he had got all the evidence he needed, he filed a case against the immigration department. The case took about three months, but Mariam and Barry were patient. Barry was confident that they would win the case. At last, they did.

Mariam was asked to go back to work and was paid her salary for the three years she had been away from the office. She happily offered to pay Barry for his services, but he refused to accept payment from her. He reminded her of her kindness to him some years earlier. He told her that he would not have become a lawyer if she had not helped him in a special way then, to speedily obtain his passport.

1. Inferencing in Pupils' Comprehension

Inferencing thereby enables the reader to create links between the information in the text and the information in the mind (Vlad, 2011). This is more real because the links the reader create in the case of inferencing are more original and authentic in comparison with what the writer would have presented if he were to connect the dots in his text. A reader armed with sufficient inferentiality would find the foregoing achievable with respect to reading comprehension. Three major subcategories of inferentiality are identified at this juncture of the analysis and they are:

- causal inferentiality in pupils' comprehension
- evidential inferentiality in pupils' comprehension.
- temporal inferentiality in pupils' comprehension

In the light of the foregoing, it is simply unrealistic to have any discourse without these three highly important aspects of inferential competence. They shall be considered in the following subsections.

Causal Inferentiality in Pupils' Comprehension

Causal inference enhances a reasoning that culminates into a conclusion that an event is or could likely be the cause of something else in the same text. Gunes (2023) observes that causal inferences are used in narrative texts to situate a sound connection between an initial situation and the problem and this goes a long way in establishing the fact that the events in a text could be an outcome of a preceding event mentioned in the text already. Competence in causal inferencing entails that a reader is able to trace events leading up to other events in a text. The following excerpt accentuates the causal inferential competence as realised in the pupils' comprehension. N.B 'Q' stands for question while 'PR' stands for pupil's response.

Excerpt 1

Question 2: Why was Mariam dismissed from her office?

PR1: Mariam was dismissed from are office for refusing to take a bribe as directed by are boss

PR2: There, he learned that she had been unlawfully dimmised.
PR3: He wanted to obtain a Nigerian passport
PR4: Mariam was dismissed from her office for refusing to take bribe as directed by her boss.
PR5: take a bribe as directed by her boss

PR1 here gave an answer that was quite apt with respect to the cause of Mariam's dismissal with the exception of the spelling error of "her" as "are" when referring to Mariam. This response viewed holistically can be said to be in congruence with the expectations of the semantic content of the answer which is the fact that Mariam refused to take a bribe as directed by her boss. PR2 presented an answer that gave a report of the unlawful dismissal of Mariam that Barry got years after the dismissal had taken place. Hence the dismissal issue was not thoroughly addressed in this response with a particular emphasis on the cause of that dismissal. PR3 obviously here presented an answer that could not trace the event that led to the dismissal of Mariam in the passage as requested by the question. This is because the Nigerian passport mentioned in the response had no direct or even indirect connection with the dismissal case. PR4 gave an appropriate response to that question by filling in the actual detail of the cause of the dismissal in the passage. PR5 supplied an answer that did a partial justice to the expectations raised by the question which is the event that led up to Mariam's dismissal. Though bribe was mentioned, the full semantic import of the answer was not addressed in this particular response.

In postulating the concept of inferential pragmatics as a section under RT, Sperber and Wilson (1995) present an approach to language understanding which centres on inferring the meaning of a communicative piece based on the evidence supplied by the writer. Evidences from the text can as well be combined with prior knowledge to come up with a more robust inferencing.

Evidential Inferentiality in Pupils' Comprehension

Evidential inferential competence as the name implies, makes use of knowledge from long term memory to process events in a text in such a way as to combine evidence in the text with prior knowledge to reach a logical conclusion. This category of inferences' processing relies on outside knowledge in order to construct a viable meaning of situations and events even while reading. Competence in evidential inference making will entail that a reader can integrate the information contained in the text with some information stored from past knowledge and experiences to create meaning. The foregoing can be exemplified in the following excerpts.

Excerpt 2

Question 6: Give two advantages of cooperation.
PR1: within a week Barry was able to obtain his passport
He thanked Mariam for her cooperation and kindness and left for the New York.
PR2: love
Kindness

PR3: The work will be done faster
The work will be effective and esier
PR4: kindness
Cooperation
PR5: helping each other
Be kindness

As a point of departure, the pupils before successfully attempting this question in the light of evidential inferential competence, must as a matter of necessity establish the fact that cooperation has advantages as can be seen in the passage. Their required task in this question was to first of all identify the fact that cooperation is advantageous. However, in addition to that, they should any supply any two of such advantages judging from how Mariam and Barry could get victory in their legal battle simply because of their cooperation. PR1 gave a response that is not in tandem with the expectations of the question. This question is actually meant to assess how the pupils could combine information from the text and their own prior knowledge to constitute an idea of what could be perceived as the benefits of cooperation. From PR1 here, we could see a direct lifting of certain portions of the passage. This particular question cannot be answered by merely lifting some portions of the passage but by a deep reflection of the benefits that cooperation can bring in the light of the legal victory recorded in the passage which is a product of cooperation between Mariam and Barry. PR2 supplied love and kindness as advantages of cooperation which is relatively factual in the light of the passage. PR3 gave a much more elaborate response than PR2 did. This is because the response is more detailed in terms of an actual conceptualisation of cooperation as being pivotal to an event like a project, hence, the response has heightened the effect of cooperation by participants in any given project by leveraging on speed and ease. PR4 gave an answer that partially accentuates the benefits of cooperation in the first part of the answer. However, the second part of the answer is just a restating of a portion of the question and as a matter of fact, the focus of the question which is cooperation. PR5 presented a response that emphasises helpfulness and kindness.

In the conceptualisation of RT, Sperber and Wilson (1995) submit that an assumption can be made about a piece of communication beyond what has been explicitly written. This assumption will be based on linguistic evidence in the text as well as prior knowledge and the combination of the two will constitute the information that has been inferred from the text based on those two streams of knowledge base.

Temporal Inferentiality in Pupils' Reading Comprehension

Temporal inferences from the word "temporal" accentuate` an understanding that a pupil has about the sequencing of events or actions mentioned in the text. In other words, temporal inferences can be regarded as the inferences a pupil will draw from the progression of activity or action in a particular chain of events after a given time as more events unfold. It can then be asserted that temporal inferential competence of a pupil can be spotted in the responses they supply to answers that index their ability to trace events after a particular period of time as required in the question. The following excerpt does quite some justice to the evaluation of temporal inferential competence in the pupils.

Excerpt 3
Question 3: What did Barry become after three years of studying in the New York?
PR1: He graduated and came back to Nigeria to attend the Law school.
PR2: Came to her office He wanted to obtain a Nigerian passport
PR3: Barry became a brilliant lawyer, known all over the country
PR4: Barry become a lawyer
PR5: Barry submitted his froms promptly

This particular question was set to assess how much the pupils could trace the events surrounding the career advancement of Barry in the passage, after his return from the New York. PR1 was merely able to state the fact that he graduated and came back to Nigeria to attend the Law school. This particular response speaks more to a means rather than an end and this is because his attendance at the Law school is not just what Barry became on his return from the New York. It was after he finished from the said Law school that he became the brilliant and famous lawyer which constitutes the essence of this question. PR2 gave a response that seemed to veer off the direction of the question and obviously showcases an inability to trace the events in the career of Barry after his years of studying Law in the New York. Hence it can be categorically asserted that PR2 presented an answer that had no semantic relationship whatsoever with the focus of the question as the substance of PR2 only spoke to the visit of Barry to Mariam's office which was long before Barry's trip to the New York.

PR3 presented an answer that did some justice to the question as it stated in unmistakable terms what precisely Barry became after his years of study in the New York which is a brilliant lawyer. There is even a major additional information that is present in PR3 which is the fame that characterised Barry as a legal practitioner. This additional information serves to enrich PR3 in terms of temporal inferential competence. PR4 supplied an answer that is relatively good because Barry actually became a lawyer after the three years of study in the New York. PR5 much like PR2 presented an answer that has no correlation with the demands of the question as there was no necessity for any forms which was not even correctly spelt.

2. Comprehension Skills' Engagement by Pupils

Comprehension skills are vital cognitive resources that a reader requires to comprehend, critically analyse and interpret texts either in written or spoken form. Highly useful for harnessing meaning, enhancing critical thinking abilities as well as mapping new information to their stipulated contexts.

Lexical-semantic Skills

These skills are highly vital to accessing word and sentence meanings as well as appropriate text interpretation. They also enable the pupils to use a wide range of words and express themselves more accurately and effectively when answering comprehension questions. Additionally, semantic skills like denotational meaning indices are essential for text understanding as to ensure an apt summarisation because the pupils would be

able to express themselves in concise and meaningful ways if they possess these skills. These skills obviously subsume the summarisation skill which serve to keep the various segments of the comprehension passage to come together as a whole and they bifurcate into two as can be seen in the following subsections.

Accessing Semantic Lexical Content

Accessing semantic lexical content of words, phrases and their interaction remains a fundamental process in text engagement which also necessitates a transformation of auditory or visual data into information that is meaningful. It also entails recognising word structures, accessing their meanings from lexical repository while at the same time integrating them into the text context. If a reader is sufficiently experienced in reading skills, such a reader would then be armed with the capacity of going through difficult texts, accessing its word meaning to construct building blocks for the text's overall meaning (Goldman, 2012). Additionally, such a versatile reader would be able to make good decisions based on such effective building blocks, thereby adjusting to the needs of cognitive and linguistic demands.

The following excerpt is from the responses of the pupils in the comprehension test scripts serve to accentuate these skills.

Excerpt 4

Q2: Why was Mariam dismissed from her office?
PR 1: She did not hear from him as he had promised.
PR 2: Barry decided that this was the time to offer his help.
PR 3: He wanted to obtain a Nigerian passport.
PR 4: because for refusing to take a bribe.
PR 5: mariam was dismissed from her office because of she wanted to help barry

DMM highlights the meaning values of words, emphasising that words act as an interface connecting phonological, syntactic, lexical, and conceptual structures during comprehension and production of utterances. Hence, the meaning values of "dismissed" index the lexical structures the pupils are to construct. PR1 responds to the question by asserting that Mariam did not hear from Barry as Barry had promised. The answer to this question is quite straightforward and requires less cognitive processing effort. Mariam was dismissed because she refused to participate in the sharp practices that were taking place at the immigration department. PR1's answer demonstrates how much he lacks simple semantic skills responsible for text meaning generation. Both PR2 and PR3 also supply answers that are semantically far from the appropriate answer. PR2 asserts that Barry decided that this was the time to reciprocate Mariam's assistance in his own dire need by helping her now. PR3 on his own also centres his own answer on the passport that Barry wanted to obtain which of course is not the focus of the question.

Only PR4 supplies an answer that is semantically accurate but grammatically incomplete. His expression passes for an adverbial clause of reason which does not express a complete thought that a sentence should express in the English language. Even though there is an instance of tautology in his response "because for refusing to take a

bribe", the expression still semantically expresses the reason for Mariam's dismissal. PR5 also gives an answer that has nothing to do with Mariam's dismissal by referring to her offer of help to Barry. Semantic skills are a precursor to pragmatic skills.

Identifying Semantic Relations

Text sentences are linked to one another through the mechanisms of meaningful interrelationships which are highly essential to constructing a meaningful text comprehension (Nahatame, 2019). Hence, semantic relationship can be understood as meaningful connections between one word or idea and another which make for a good text interpretation. Such semantic relationships include synonymy, antonymy, and polysemy to mention a few. For want of space, we shall be considering the semantic relation of synonymy and the following excerpt would exemplify the concept.

> **Excerpt 5**
> Question 5: Were Mariam and Barry anxious or impatient during the case?
> PR1: Barry was condifient that they would win the case
> PR2: Mariam and Barry anxious or impatient during the case
> PR3: Barry was confident that they would win the case
> PR4: The case took about three months but Mariam and Barry were patient.
> PR5: Barry was confident that they would win the case at last they did.

In line with the postulations of RT, when a message is communicated, recipient combines the information in the message with their expectations about relevance, prior knowledge, and sensory input to infer the communicator's intended meaning. "Anxious" in the fifth question shares a semantic relationship with "impatient" in the context of this question which is known as synonymy. This implies that both can be used interchangeably as it is in this particular context and it means to be patient and allow things run in their normal course instead of manipulating one's way out. The pupils are therefore supposed to combine their prior knowledge about the semantic relationship between the two words. PR1 gives a response that is not far from the correct response as it gives a hint on the fact that Barry was confident that they would win the case but did not include the patient disposition of Mariam since the question was asked about the two main characters in the story, Mariam and Barry. PR2 presents an answer that merely restated the question without a reasonable attempt to answer the question satisfactorily. PR3 presents an answer that just like PR1 only paid attention to Barry's emotional state during the three months that the case lingered in the court of law. PR4 presents a response that did quite some justice to the essence of the question. In the passage, one could see explicitly the patience of both Barry and Mariam and this particular response pointed that out quite plainly. PR5 also gives an answer that also emphasises the patience of Barry alone but then added the fact that at last they won the case.

The pupils will undoubtedly have to recognise the synonymous relationship that "anxious" shares with "impatience" in order to be able to do justice to this particular question.

Pragmatic Comprehension Skills
These skills are functional as they foreground the pupils' ability to conceptualise those meanings in structures that are not clearly stated in words but which are supposed to be understood during the course of reading. These skills are very crucial as they underscore the need for pupils to be able to read beyond the author's penned down content. Some subtle messages are passed across from the writer to the reader which must be pragmatically decoded. Pragmatic skills are therefore subcategorised as can be seen in the following subsections.

Contextualising meaning in pupils' comprehension
Contextualising meaning has to do with situating meaning in its appropriate situational context so as to pin meanings down to specific and tangible pragmatic values. Contextualising meaning in comprehension also arms readers with sufficient pragmatic resources to disambiguate expressions properly in instances of ambiguities when there are multiple meanings for a single expression. Additionally, contextualising meanings would enable an in-depth interpretation and overall understanding of text as well as make information retention possible. Making use of contextualied reading materials serve to motivate pupils to individualise learning by engaging their schema by connecting text to its context thereby boosting readers' confidence (Gerundio, 2023). The following excerpt serves to expatiate this point further.

> **Excerpt 6**
> Q4: How did Mariam and Barry work as a team?
> PR1: Barry approached Mariam and told her of his great need.
> PR2. He thanked Mariam for her cooperation and kindness
> PR3: Mariam lived in Zaria she worked at the immigration office
> PR4: Barry and Mariam decided to work together as a team.
> PR5: Mariam supplied him with all the information he needed and he noted all the illegal deal that took place in the immigration department.

Van Dijk's (2009) subjective theory of context establishes the fact that mental models also conceptualised as context models are constructed by participants in communicative interaction. These models go a long way in determining how well a speech situation is understood against the backdrop of the current realities in the communication exchange. However, a proper understanding of such current realities is key even in such a subjective context. In the light of the provisions of the tenets of Van Dijk's context theory, a correct conceptualisation of "work" is very important here in this very question as well as the accurate contextualisation. This is because the "work" here differs from the regular job that people take to get paid at the end of a week or the end of a month. It speaks more of a sort of collaboration between both Barry and Mariam in engaging in a legal tussle over the illegitimate dismissal of Mariam from the Immigration department. PR1 gives an answer that leverages on the first meeting of both Barry and Mariam and the expression of Barry's great and urgent need without a single mention of the sort of their togetherness. PR2 presents a response that spotlights the gratitude of Barry to Mariam for the latter's cooperation and kindness. This response also did not do justice to the examiner's

expectation on that question. PR3 presents a response that shows a big misconception of the sense of the "work" in this particular question. The concept of a 9-5 work has been mistaken for a togetherness embedded in two or more people working together with unity of mind. PR3 referred to the work that can contextually replace "job", hence the supply of a wrong answer. PR4 presents an answer that merely restates the content of the question with the exclusion of the question word "how". PR5 gives a response that shows a contextually appropriate sense of the concept of "work" as mentioned in the passage. Additionally, the response adequately points out the core essence of the togetherness of Barry and Mariam which is the fact that Mariam supplied Barry with all the information he needed to make a reasonable case before a judge in a court of Law while he on his part he noted all the illegal deals in the immigration department.

As indicated in the foregoing, quite a number of the pupils examined lack pragmatic skills which could serve as contextual cues to recover meanings of unfamiliar words or sentences.

Implicatures' Retrieval in Comprehension

Implicatures are information not explicitly stated in a text but can be inferred. In some other cases, implicatures may engender some other facts not necessarily mentioned in the fact but can be understood as consequences or results. In the case of Second language learners, coming to terms with these implicatures can be tasking due to particular linguistic as well as cultural differences between the native language and the target language. The following excerpt exemplifies the importance of retrieving contextual meaning in comprehension.

Excerpt 7

Q6: Give two advantages of cooperation.
PR1: we should help each other.
PR2: and left for the new york
PR3: kindness to him some years earlier
PR4: you must cooperate what the teacher has written
PR5: she show love to him

The foregoing responses reveal that the pupils are not armed with sufficient pragmatic skills which should enable them do justice to the above question. The theme of cooperation pervades the entire passage, hence a pupil with pragmatic skills should be able to come up with advantages of cooperation through the lens of the passage. Even though the advantages of cooperation were not literally written in the passage, a reader with inference skills which pragmatics can afford, should be able to identify specific advantages of cooperation because the fact that both Mariam and Barry cooperated in the passage led to a successful legal battle. PR1 writes that "we should help each other" which though not far from the answer, is still not the accurate answer. PR2 and PR3 present expressions that could be considered sentence fragments which are not accurate answers to the question. PR4 responds to the question by bringing up an issue on cooperating with what the teacher has written in which case no mention was made of any teacher all through the entire passage. PR5 hinges his answer on love which is not even the focus of

the passage. Sperber and Wilson (1995) in their conceptualisation of RT visualise pragmatic comprehension as the ability to infer the communicative intention of the writer or speaker as the case may be. Inference in the provisions of RT is quite applicable here as it bifurcates into explicatures and implicature. The explicatures are expressions that are explicitly stated in a communication piece while implicatures are conveyed without actually stating them literally and this is the core of inferencing. Inferencing is achieved by making use of contextual cues presented by the speaker or writer. In this case, these pupils could not infer the communicative intention of the examiner in this particular question which is to infer advantages of cooperation based on the success story of Barry and Mariam in the legal battle consequent upon their cooperation.

Discourse Comprehension Skills

These skills involve building meaning structures from extended segments of language, entail successfully comprehending larger units of texts, require making inferences to connect ideas both within and across local and global discourse contexts, and, following successful comprehension, allow the resultant discourse representation to be retrieved, updated, and applied in order to answer questions and solve problems. An absence of these skills however, is tantamount to a breakdown of comprehension because comprehension texts are presented in a discourse mode since they are mostly large stretches of language as opposed to a single-sentence text. Kecskes (2008) in his Socio-cognitive Approach situates the actual situational context just in-between the lexical and conceptual levels. The implication of that is that a reader will have to extensively rely on the context which is encapsulated in the words of the specifically chosen by the writer in order to build author's intended meaning.

These discourse skills are then germane as they will serve to enable the pupils to build models of meaning which they should be able to achieve by sourcing the actual situational context from the lexical choices of the writer. These particular skills are responsible for how a reader can monitor self-comprehension. Discourse skills are further subcategorised into two as can be seen in the following subsections.

Accessing Local Discourse Comprehension

Local discourse comprehension in reading comprehension resonates with the cognitive process of making sense of the interrelationships between closely constructed sentences and small text segments so as to build a coherent meaning on a combination of the sense in those sentences. Kucer (2013) assert that expressions read in a way that maintains the author's intended meaning notwithstanding the presence of miscues are positively correlated with local level of comprehension. This has implications for sourcing for an author's intended meaning to arrive at a local level of comprehension. The following excerpt accentuates this fact.

Excerpt 7

Q1: What help did Mariam give to Barry?
PR1: Mariam lived in Zaria
PR2: Barry came to her office He wanted to obtain a Nigeria passport
PR3: He wanted to obtain a Nigerian passport

PR4: obtain a Nigerian passport and to make it worse, he needed the passport urgently
PR5: She give him the application forms and to fill them.

van Dijk (2008) in his SCA, affirms that for a reader to understand a language piece, he should reconstruct the communicative intention of the speaker. PR1 gives an answer that showcases a shallow comprehension of the initial part of the comprehension passage as the response did not do justice to the core of the help that Mariam rendered to Barry. PR2 gives an answer that hinted his need that he presented to Mariam. The answer only did not include the fact that Mariam actually assisted him to process that Nigerian passport to time. PR3 also mentioned the fact that he wanted to obtain a Nigerian passport but just like PR2, the answer was not detailed enough to do justice to the demands of the question. PR4 gives an answer that leverages on the urgency of the need for Barry to secure his passport within just a week. However, the response was a better attempt at satisfying the demands of the question. PR5 presents an answer that stops at the giving of the application forms to Barry and the fact that he filled them, leaving the other efforts of Mariam out of the answer.

In actual fact, the answer to this question does not have to span across the entire text unlike the global discourse comprehension that may encompass an understanding of a good part of the text if not the entire text.

Accessing Global Discourse Comprehension

Global discourse comprehension hinges on the overall theme of a text. This is in sharp contrast to the local discourse comprehension level which focus on few text segments as the global discourse level encompasses the entire text's meaning. Recent psychological theories have tried to come up with the identification of deixes, the connection of text segments that are explicit as well as the establishment of global coherence (Graesser, Millis and Zwaan, 1997). This is to underscore the vital elements that make for a successful global discourse comprehension. The following excerpt from the responses of the pupils in the comprehension test scripts amplify the importance of global discourse comprehension.

Excerpt 8

Q4: How did Mariam and Barry work as a team?
PR 1: he filed a case against the immigration department
PR 2: Mariam was asked to go back to work and was paid her salary
PR 3: Barry decided that this was the time to offer his help
PR 4: Barry and Mariam decided to work together as a team.
PR 5: Because of illegal deals that took place in the immigration department

In line with the provisions of RT, meaning should be recovered from the producer's perspective. PR1 spotlights the activity of Barry alone without making any reference whatsoever to the activities of Mariam. This singular act of PR1 demonstrates poor discourse skills as he could not connect different segments of the passage pointing to the togetherness of Barry and Mariam on the legal tussle. PR2 gives an answer that is

semantically distant from the required answer. His own answer hinges on the aftermath of the legal case which is the verdict that Mariam should go back to work and should be paid her salaries including the ones she did not receive all through the period of her suspension. PR3 makes his own answer to reflect the event that led up to how the case began because it all began when Barry understood the huge injustice done to Mariam at the immigration department. It was then he now decided to help Mariam as a return for her kind deed in helping him to secure his passport to time which then enabled him to travel to the New York to bag his law degree in the first place.

PR4 merely restated the question as his own answer. This form of response demonstrates a poor comprehension ability as he could not even create any coherent model of understanding using the different segments of the comprehension text. PR5 answered the question in a way that shows part of the reason for the legal case in the first place which happens to be the shady deals frequently occurring at the immigration department. Kroeger (2017) submits that most questions specify the kind of information that is expected of the addressee to supply. The foregoing fact is explicitly captured in the concept of the question-answer congruence test which is a subcategory under focus in information structure. The way these pupils connect pieces of information in their answers showcases their current cognitive status like their knowledge base. Their lexical choices reflect their level of understanding of discourse stretches.

Conclusion

The study set out to investigate Nigerian pupils' ability to engage inferentiality and cognitive-pragmatic skills to tackle reading comprehension task. Having established the fact that inferencing and cognitive-pragmatic skills are essential requirements for in-depth reading comprehension to take place, Nigerian pupils should therefore be immersed in adequate and systematic instructions with the intent of inculcating these essential skills in order for them to excel academically since effective reading comprehension ability is a major precursor to academic achievements in school subjects.

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